Jaehwan Jeong

Email | Google Scholar | LinkedIn | Website

Research Interests

Embodied AI, Robotic Learning, Autonomous Systems, Sim-to-Real
Vision-Language Model, Multi-modal learning, Generative model, AI safety

Education

- **Korea University** — *Mar. 2024 - Fe. 2029 (Expected)*
  Ph.D. - Artificial Intelligence — Seoul, South Korea
- **Chung-Ang University** — *Mar. 2017 - Fe. 2021*
  B.E. - Electrical & Electronic Engineering — Seoul, South Korea

Experience

- **Computer Vision Lab @ Korea University** — *Mar. 2024 - Present*
  Ph.D. Student Researcher (Advisor: Prof. Sangpil Kim)
  - Adversarial noise for Deepfake AI safety *(collaborated with Dr. Jaewook Chung @ **SAMSUNG Research**)* - **[C2]**
  - Multi-modal audio-to-video generation *(collaborated with Dr. Eugenio Culurciello @ **PURDUE Univ.**)* - **[J1]**
- **Structures-Computer Interaction Lab @ University of California, Los Angeles (UCLA)** — *Mar. 2025 - Mar. 2026*
  *Visiting Graduate Researcher; **3D Farm Project Lead** (Advisor: Prof. M. Khalid Jawed)*
  - VLA & RL-driven Sim2Real deployment for mobile robots *(collaborated with Dr. Joo @ **NVIDIA Research**)* - **[C3]**
  - 3D Plant skeletonization & Physics-based simulation for robotic pollination - **[J3]**
  - Robotic agricultural 3D reconstruction benchmark and dataset *(collaborated with **NDSU**)* - **[J2]**
- **Computer Vision Lab @ Korea University** — *Jul. 2023 - Fe. 2024*
  Undergraduate Research Intern (Advisor: Prof. Sangpil Kim)
  - Diffusion-based long video generation *(collaborated with Dr. Byeon @ **NVIDIA Research**)* - **[C1]**
- **Military Officer** — *Mar. 2021 - Jun. 2023*
  Signal Company, 5th Armored Brigade, Republic of Korea Army
  - Operations and Tactical Planner - 1st Lieutenant
  - Wired communication network management (UTP, Optical cables) - 2nd Lieutenant

Publications

C=Conference, J=Journal

[C3] **Jaehwan Jeong**, Evelyn Zhu, Jinying Lin, Emmanuel Jaimes, Tuan-Anh Vu, Jungseock Joo, Sangpil Kim, M. Khalid Jawed, ***Your Vision-Language-Action Model Already Has Attention Heads For Path Deviation Detection***, under review, 2026 [PDF]

[J3] **Jaehwan Jeong***, Tuan-Anh Vu*, Radha Lahoti, Jiawen Wang, Vivek Alumootil, Sangpil Kim, M. Khalid Jawed, ***Vision-Guided Targeted Grasping and Vibration for Robotic Pollination in Controlled Environments***, under review, 2026 [PDF]

[J2] **Jaehwan Jeong**, Tuan-Anh Vu, Mohammad Jony, Shahab Ahmad, Md. Mukhlesur Rahman, Sangpil Kim, M. Khalid Jawed, ***AgriChrono: A Multi-modal Dataset Capturing Crop Growth and Lighting Variability with a Field Robot***, under review, 2026 [PDF] [Project] [Code]

[C2] **Jaehwan Jeong**, Sumin In, Sieun Kim, Shin han yi, Jonghen Jeong, Sang Ho Yoon, Jaewook Chung, Sangpil Kim, ***FaceShield: Defending Facial Image against Deepfake Threats***, International Conference on Computer Vision (**ICCV**) 2025 [PDF] [Code]

[J1] Sieun Kim, **Jaehwan Jeong**, Sumin In, Seung Hyun Lee, Seungryong Kim, Saerom Kim, Wooyeol Baek, Sang Ho Yoon, Eugenio Culurciello, Sangpil Kim, ***Semantically Complex Audio to Video Generation with Audio Source Separation***, Engineering Applications of Artificial Intelligence (**JCR IF Top 10%**) 2025 [PDF] [Code]

[C1] Gyeongrok Oh, **Jaehwan Jeong**, Sieun Kim, Wonmin Byeon, Jinkyu Kim, Sungwoong Kim, Sangpil Kim, ***MEVG: Multi-event Video Generation with Text-to-Video Models***, European Conference on Computer Vision (**ECCV**) 2024 [PDF] [Code]

Academic Service

- **Conference Reviewer**
  CVPR'25-26, ICCV'25, ECCV'26, NeurIPS'26, ICLR'26, BMVC'26, DLT'25
- **Journal Reviewer**
  EAAI'25-26, CVIU'25, NN'26, TMC'26, PR'26

Patents

[P1] **Jaehwan Jeong**, Sangpil Kim, ***Method and Apparatus for Protecting Facial Image based on Disturbance Signal to Counter Deepfake Attack***, Korean Patent, No. 10-2025-0118588 (2025.08.25)

Skills

- **AI & Perception**
  Vision-Language Models, Reinforcement Learning, Adversarial Attack, Diffusion Models, GANs
- **Robotics & Sensor Fusion**
  Unmanned Ground Vehicle (UGV), Robotic Arms, Simulation, ROS 2, RGB-D Cameras, LiDAR, IMU

References

- **Assoc. Prof. Sangpil Kim**
  Department of Artificial Intelligence, Korea University, South Korea
- **Assoc. Prof. M. Khalid Jawed**
  Department of Mechanical and Aerospace Engineering, University of California, Los Angeles
- **Dr. Jungseock Joo**
  Senoir Engineering Manager @ NVIDIA | ex-Prof @ UCLA
- **Postdoc. Tuan-Anh Vu**
  Department of Mechanical and Aerospace Engineering, University of California, Los Angeles